

مصرف

A Finance Decision Assistant

CSC 440 : Human-Computer Interaction
Course Project

Prepared by:

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TABLE OF CONTENTS

<i>Introduction:</i>	3
<i>Problem definition:</i>	3
<i>target users:</i>	3
<i>research findings</i>	4
<i>functional requirement:</i>	7
<i>usability requirement</i>	7
<i>persona</i>	8
<i>task flows</i>	9
<i>wireframes</i>	10
<i>prototype explanation</i>	13
<i>usability testing results</i>	20
<i>improvements</i>	20
<i>Appendix 1 – questionnaire questions</i>	22
<i>Appendix 2 – Responses of the opened question (question 9)</i>	23
<i>Figma url</i>	23
<i>AI acknowledgment</i>	23

INTRODUCTION:

Managing personal finances is one of the main challenges faced by university students today. With a limited monthly allowance and many daily small expenses such as coffee, meals, transportation, and subscriptions, students often struggle to keep track of their spending, which may lead them to run out of money before the end of the month. Traditional expense-tracking methods are usually time-consuming and easy to forget during a busy academic schedule, while many existing financial applications are too complex for students' needs. Therefore, this project aims to design a simple and user-friendly financial management application specifically for university students. By applying Human-Computer Interaction (HCI) principles, the application focuses on usability, simplicity, and accessibility to help students track their expenses more effectively and develop better financial habits.

PROBLEM DEFINITION:

The main problem addressed by our application is that tracking personal expenses requires users to manually enter their spending data, which is often a tedious and time-consuming task. Many individuals find it difficult to consistently record every transaction, especially in their busy daily lives. As a result, users frequently forget or neglect to log their expenses, leading to incomplete or inaccurate financial records. This lack of reliable data makes it challenging for individuals to monitor their spending habits, make informed financial decisions, and maintain effective budgeting. Therefore, there is a clear need for a solution that simplifies or automates expense tracking, reducing user effort while improving accuracy and consistency.

To address these challenges, the application is designed with several design goals. First, the system aims to provide a simple and user-friendly interface that allows users to record and access financial information quickly and easily. Second, the application seeks to minimize manual effort by supporting automated or simplified expense entry methods. Third, the design focuses on improving accuracy and consistency in expense tracking to help users maintain reliable financial records. In addition, the application aims to present financial data through clear visualizations and spending analysis, enabling users to better understand their financial habits. Another important goal is to support effective budgeting by providing alerts, insights, and categorized expense monitoring. Finally, the system is designed to ensure accessibility, responsiveness, and an overall smooth user experience across different devices.

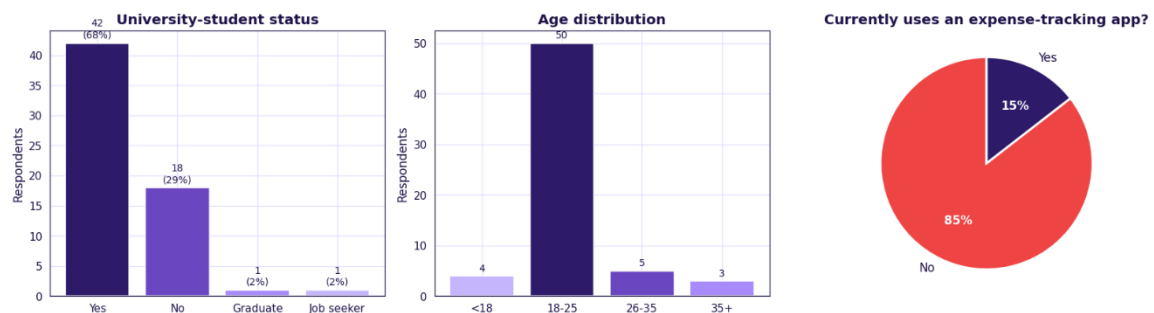
TARGET USERS:

The primary target users of this application are university students. Students often manage limited budgets while balancing academic responsibilities, social activities, and personal commitments. Due to their busy schedules, they may struggle to consistently track their expenses, which can lead to poor financial management. This application aims to support students by providing an easy and efficient way to monitor their spending and make better financial decisions.

RESEARCH FINDINGS

For this project, we have conducted an 11-question questionnaire (questions found at *appendix 1*), distributed the questionnaire through student channels to gather insights, and acquired 62 responses. questionnaire conducted to profile the respondent base, Quantify current behavior around expense tracking, Measure pain points of their tracking journey (manual entry, forgetting, chart confusion), Rank desired features, and nine open-ended pain-point responses for recurring themes. This questionnaire was conducted to understand the current state of the problem and to direct an appropriate user experience design.

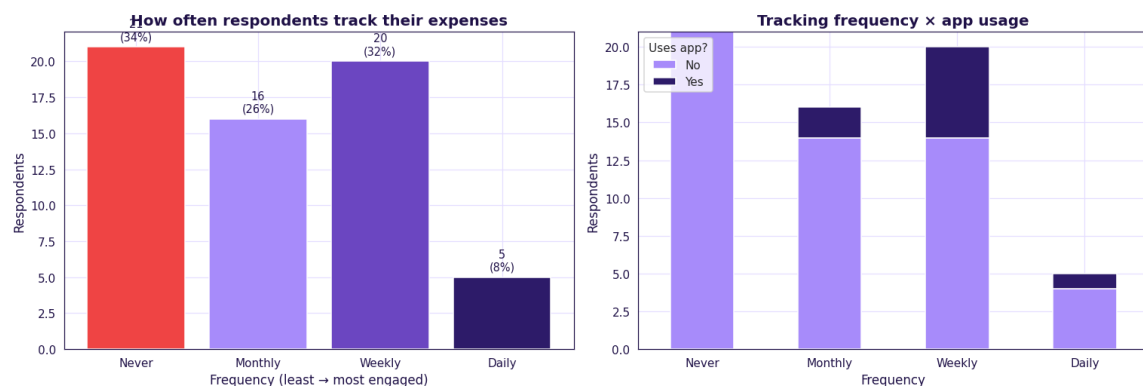
To understand the respondent base, we asked if the person is a student or not, their age group and if they are currently using a tracking app or not. And the findings are displayed in the graph below.



In short, the sample skews heavily young and student-leaning, where 68% are university students, 81% (50) are within 18-25 age group. And only 15% of the respondents uses an expense tracking app.

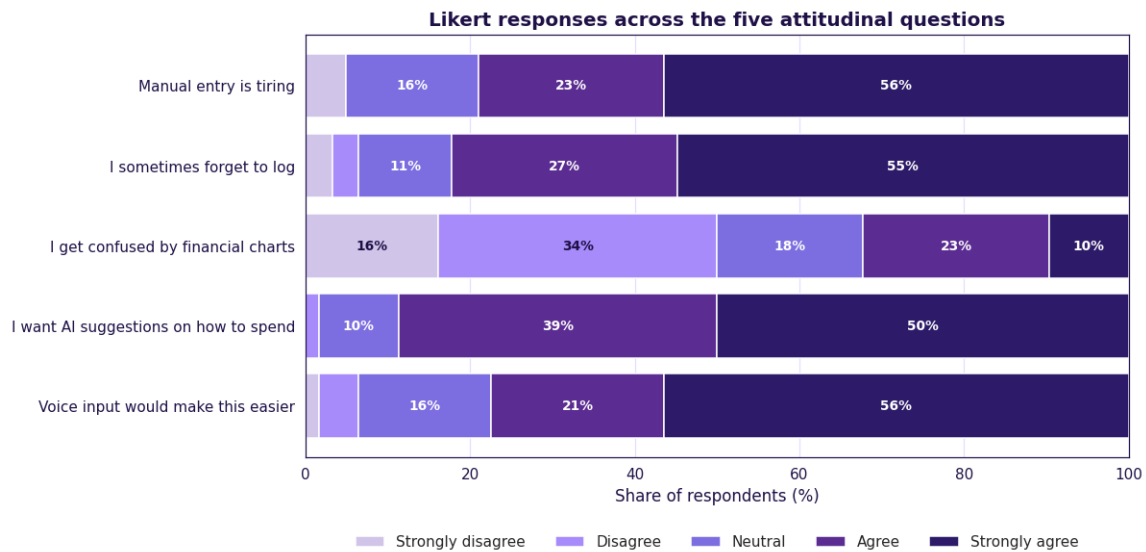
We chose university students at age 18–25 as our target user based on their limited financial income and their early life stage, where they need to adapt and learn financial management skills.

To interpret how the respondents tracking behavior are we asked a question about their tracking frequency and the results displayed in the below figure.



21 of 62 respondents (34%) report that they never tracks their expenses, whereas only 8% percent track their expenses daily. And Among the 53 non app users, 32 still track there expenses frequently (weekly or monthly).

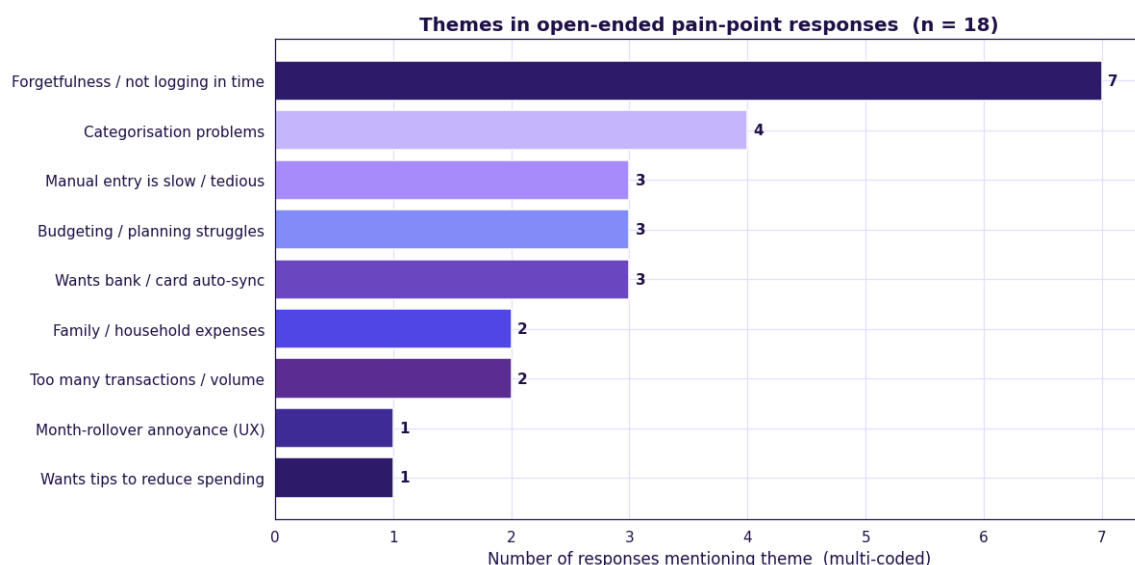
Next, we asked 5 behavioral financial tracking question in Likert scale (where we represent a behavior and the respondents rate their level of agreement or disagreement with that behavior), the 5 behavior and their related results are displayed in the next figure.



By interpreting the results, we found that almost 79% agree or strongly agree that manual entry is tiring, and 82% agree or strongly agree that they forget to log. And only 32% of target users feel confused by charts.

These results helped as a lot in the designing part, where we set our focus to make the entry of the expenses easier, and developing ways to remind the user to log their expenses. And the strong results are combined with strong agreement on voice input, indicating the expectations and hopes from such a modern app design. And since the low percentage of respondents feels confused by the charts, we made sure that our charts are generally simple but did not heavily invest on that.

And for the open-ended question, we. Asked about problems they face when tracking their expenses, as we made answering this question optional, only 18 respondents provide answers (answers can be found at *appendix 2*). The results after mapping them to themes are as displayed below in the figure.

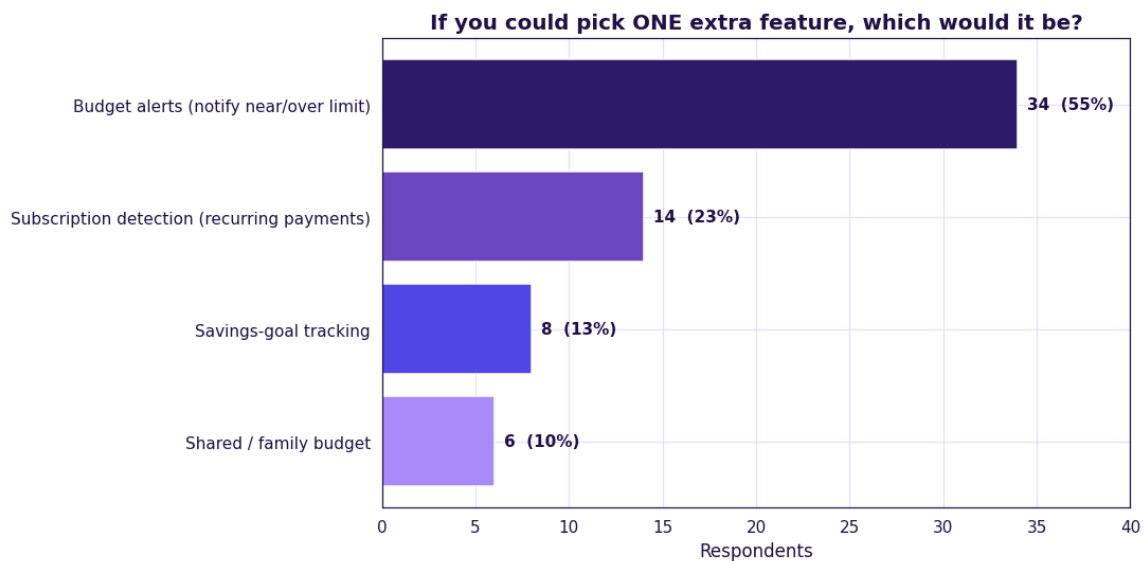


The three biggest themes in the above responses are forgetfulness, slow manual entry, and the desire for bank/card auto-sync. This perfectly mirrors the previous findings and our goals in reducing how the users forget, how they find entry tiring, and how to provide more automatic solutions.

In addition to these, one respondent flagged the month-rollover behavior as annoying, as the app their using, moves them to a new month before they've finished entering the old one. In our designing process, we are flagging these issues in mind to make the app more usable and reliable.

For our last question, we asked the user about **“If you could choose one additional feature, which would you pick?”**, and addressed 4 possible features.

The result found in the figure below, shows that 55% of the respondents choose budget alert on the rest of the features, and the second wanted is subscription detection. Whereas the least wanted are saving goals tracking and share gudgeting feature.



In conclusion, the questionnaire results helped in pointing clearly toward a clear design direction. Within our target users of Saudi university students aged 18–25, the dominant barriers to expense tracking are the friction of manual entry and the tendency to forget logging transactions. The path forward is to minimize entry effort through modern features like voice input, and to engage the user through budget alerts as they were the most-requested feature at 55%.

FUNCTIONAL REQUIREMENT:

- **The system shall allow users to create an account and log in.**
- **The system shall allow users to add, edit, and delete expenses.**
- **The system shall categorize expenses (e.g., food, transport, bills).**
- **The system shall allow users to add expenses using voice notes.**
- **The system shall process voice input and convert it into structured expense data.**
- **The system shall allow users to view spending trends over time (e.g., weekly and monthly).**
- **The system shall allow users to set and manage personal budgets.**
- **The system shall send budget alerts when the user is approaching or exceeding their set budget.**
- **The system shall analyze spending patterns to generate personalized recommendations.**
- **The system shall provide financial suggestions based on the user's spending behavior.**

USABILITY REQUIREMENT

- **Students must be able to complete adding an expense in 3 clicks.**
- **New users should be able to set a monthly budget for the first time without needing any help or instructions.**
- **The spending trends must be clear enough for a student to understand their financial status within 3 seconds of looking at the screen.**
- **The system must provide an "Undo" button after deleting any record to prevent mistakes and reduce user frustration.**
- **Budget alerts must use motivating and friendly language to encourage students rather than just showing warnings.**
- **The system must show a confirmation message before deleting any financial data to avoid accidental loss.**

PERSONA

PERSONA 1:

BACKGROUND

- 21, Female
- University Student (Computer Science)
- Receives monthly allowance
- Comfortable using smartphone & apps

MOTIVATIONS

- Make allowance last until the end of the month
- Save money for goals
- Become more financially responsible

FRUSTRATIONS

- Manual entry is the single biggest source of pain
- Forgetting to log on the day, then having to reconstruct from memory
- Generic categories that don't fit how she actually spends

NEEDS

- Know where her money actually goes each month
- Weekly insights and spending trends
- Voice input
- Specific sub-categories

Sara

Sara is a university student who depends on her monthly allowance to manage transportation, food, coffee, subscriptions, and daily expenses. She often spends small amounts throughout the day without realizing how quickly they accumulate.

"Adding expenses manually is tiring, takes time, and I forget."

"I forget to write it down. I don't stick to the plan."

"I forget what I spent on."

"I hate that the app moves me to a new month before I've logged the previous one's expenses, and it doesn't have clear enough categories, like if I'm spending on myself, it should have sub-categories: skincare, vitamins, so I know exactly where the money went."



PERSONA 2:

BACKGROUND

- 20, Male
- Mechanical Engineering Student
- Receives monthly university allowance
- Tech-savvy and uses multiple digital services
- Lives in university dormitory

MOTIVATIONS

- Manage allowance efficiently
- Build savings for gadgets and personal projects
- Invest in skills and courses
- Achieve financial independence in the future

FRUSTRATIONS

- Makes impulsive purchases
- Spends a lot on gaming and online subscriptions
- Doesn't plan his budget in advance
- Hard to resist discounts and online deals
- he knows he's not doing the thing he should

NEEDS

- Budget planning and limits
- Alerts to control impulsive spending
- Alerts that come to him, he won't open the app proactively
- A reason to care, alerts that feel useful, not noisy

Faisal

Faisal is a university student who receives a monthly allowance from the university. He enjoys technology, gaming, and trying new gadgets. He often spends on things he wants in the moment without planning ahead, and has never seriously tried to track expenses. Doesn't feel acute financial pain, just a vague unease about where his money goes.

"Every day I buy things like coffee, food at university, online orders, I don't feel I have time to sit and enter everything as it takes so long."

"I spend before I think, and at the end of the month I always regret it."

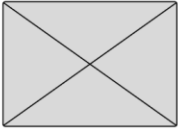


TASK FLOWS

Task Flow 1: Add an Expense	Task Flow 2: View Spending Trends
Goal: Allow students to quickly record daily expenses with minimal effort. Main Flow: 1. User opens the application. 2. User logs into their account. 3. User taps the "Add Expense" button from the home screen. 4. User chooses input method: • Manual entry • Voice input 5. User enters and chooses: • Amount • Category • Date (optional auto-detected) • Note (optional) 6. System automatically categorizes the expense if voice input is used. 7. User taps "Save". 8. System confirms successful addition. 9. Updated balance and spending summary appear on dashboard. Alternative Flow: • If voice recognition fails: 1. System shows friendly error message. 2. User can retry voice input or switch to manual entry.	Goal: Help students understand their spending habits clearly and quickly. Main Flow: 1. User opens the application dashboard (Home). 2. User selects "Spending Trends". 3. System displays: • Daily spending chart • Weekly spending chart • Monthly spending chart • Yearly spending chart • Total income and expenses for all categories 4. User filters data by: • Days • Week • Month • Year AND/OR • One or more expense categories 5. If the user selects one or multiple expense categories, the system displays spending data and category breakdowns related only to the selected categories. 6. The user can view the selected categories' data based on: • Daily trends • Weekly trends • Monthly trends • Yearly trends 7. System instantly updates: • Charts • Income totals • Expense totals • Spending analytics 8. User reviews highest spending categories, financial habits, and spending trends. Alternative Flow: • If there is insufficient data: System displays a friendly message: "Start adding expenses to view your spending insights."
Task Flow 3: Receive Financial Suggestions	Task Flow 4: Budget Alerts
Goal: Provide personalized recommendations to improve financial habits. Main Flow: 1. User opens the dashboard. 2. System analyzes recent spending behavior. 3. User selects "Financial Suggestions". 4. System displays personalized recommendations such as: • "You spent 30% more on coffee this week." • "Consider reducing food delivery expenses." • "You are staying within your monthly budget." 5. User can tap any suggestion to view more details or tips. Alternative Flow: • If not enough spending data exists: System asks the user to add more expenses first.	Goal: Notify students when they are close to exceeding their budget. Main Flow: 1. User opens "Budget Overview". 2. User sets a monthly budget amount. 3. System monitors spending automatically. 4. When spending reaches a threshold (e.g., 80%): • System sends a friendly alert notification. 5. User opens the notification. 6. System shows: • Current spending • Remaining budget • Suggested actions to reduce spending Alternative Flow: • If user exceeds budget: 1. System sends warning alert. 2. System highlights categories causing overspending.

WIREFRAMES

1- Log in & Sign up:



مصرف

Log In

Sign Up

Forgot Password?

Welcome

Username Or Email

Password

Log In

Sign Up

Create Account

Name

Email

Mobile Number

Password

Confirm Password

Sign Up

Already have an account? Log In

2- Home page and add expenses:

Hi, Welcome Back

Total Budget: \$990.00 | Total Expense: ~\$460.40

Lastest Transaction

Saving	-\$100.00
Groceries	-\$100.00
Rent	-\$260.40

+ Microphone

Home, Budget, Add Expense, Transactions, Profile

Add Expenses

Select The Category

+ Category

Add Expenses

Food

Amount* 25.00

Date* Today Yesterday

April 30 2028

Time Evening Morning Noon Midnight

Or specify 12:10 AM PM

Method Apple Pay Name Coffee

Add Expenses

Expense Added Successfully

Back to Home

Notification

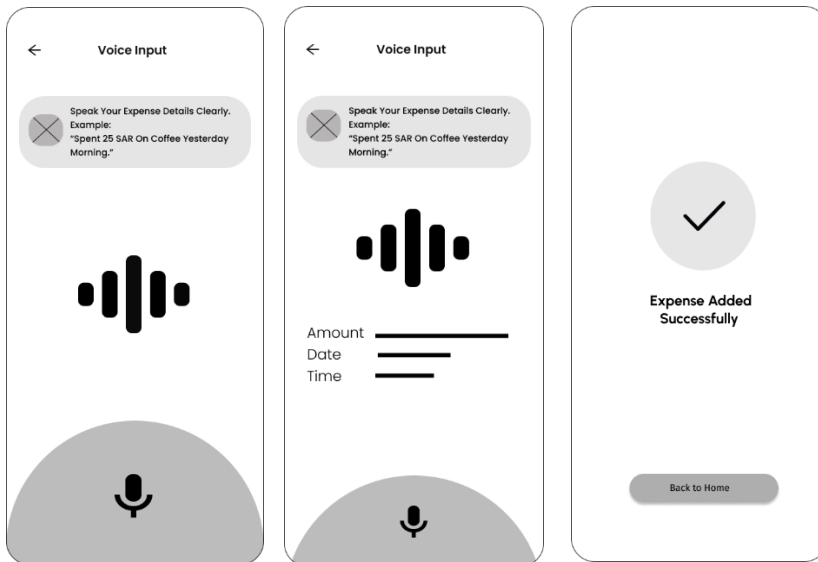
Today Budget Alert

Smart Suggestion

Yesterday Savings Goal Reminder

This Weekend Achievement Notification

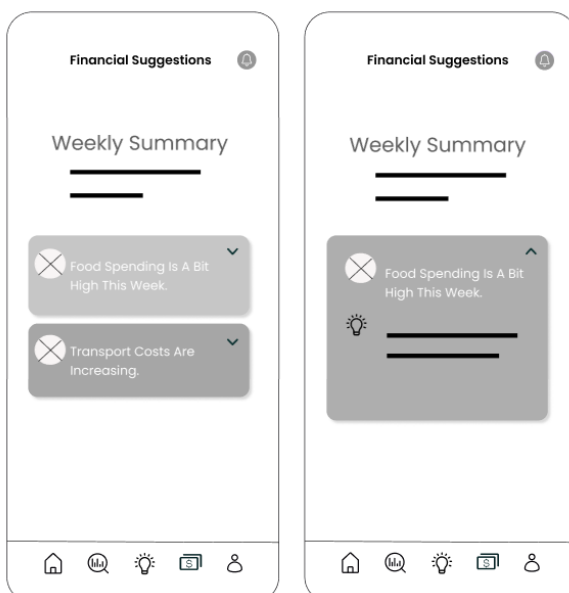
2.2-Add expenses by microphone



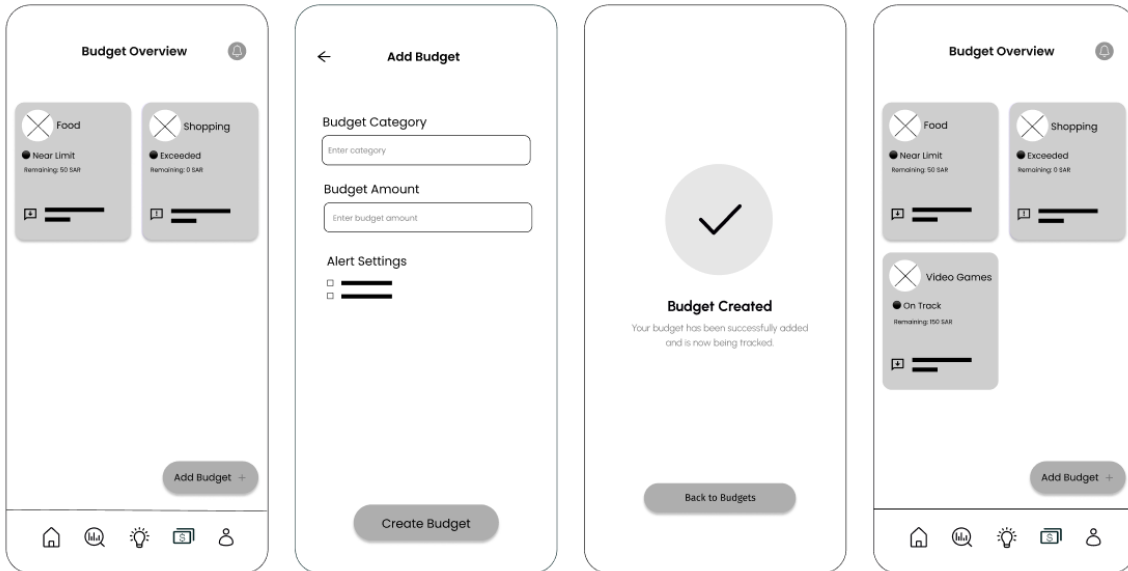
3- Spending Trends page:



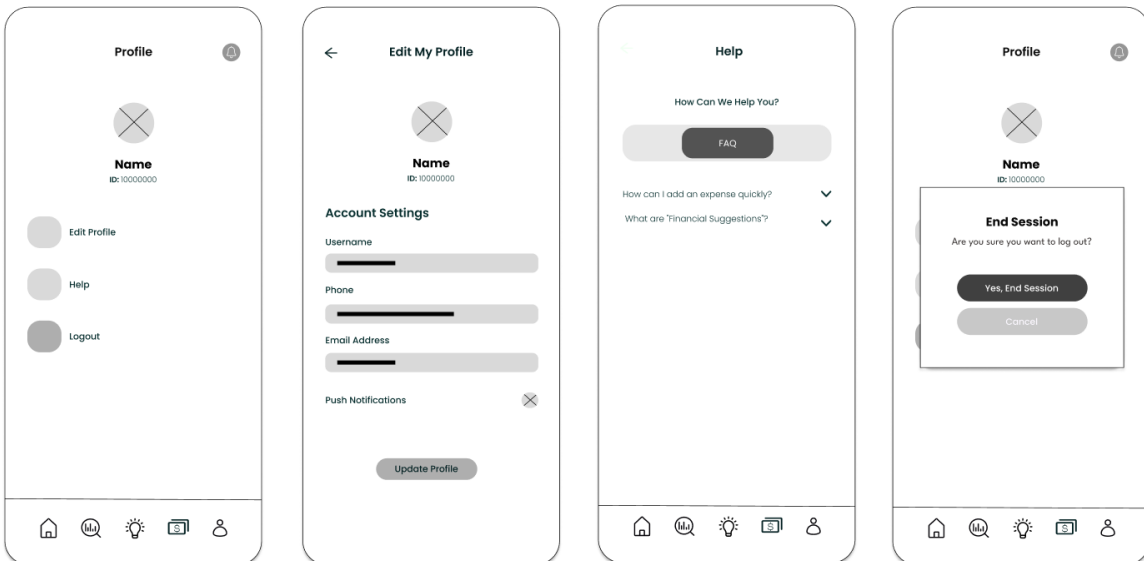
4- Financial Suggestions page:



5- Budget page

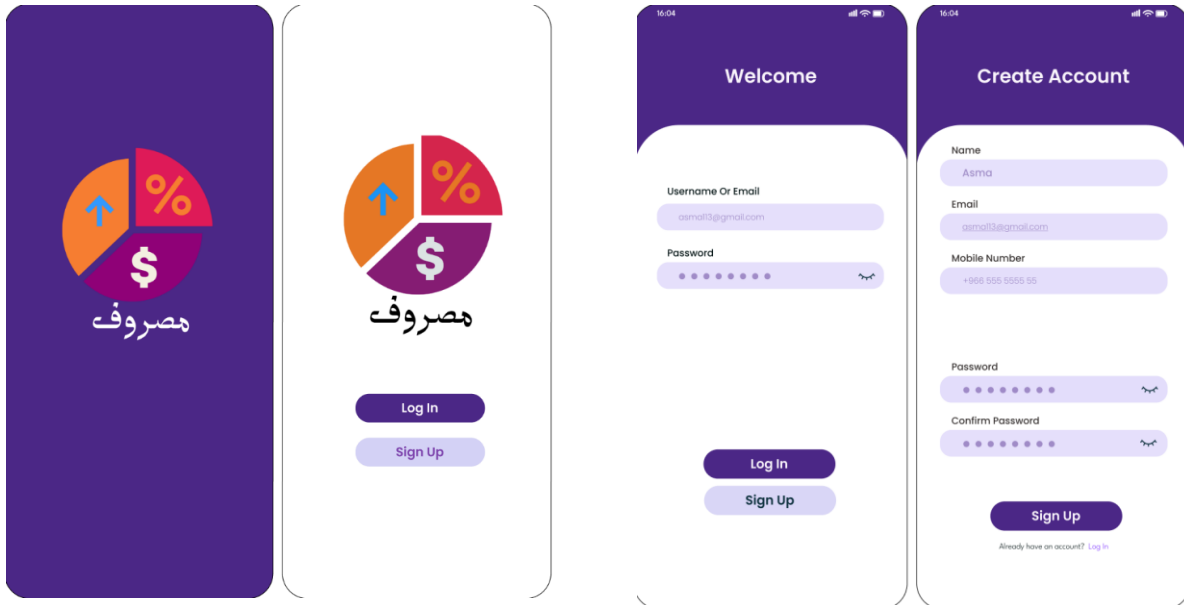


6- Profile page:

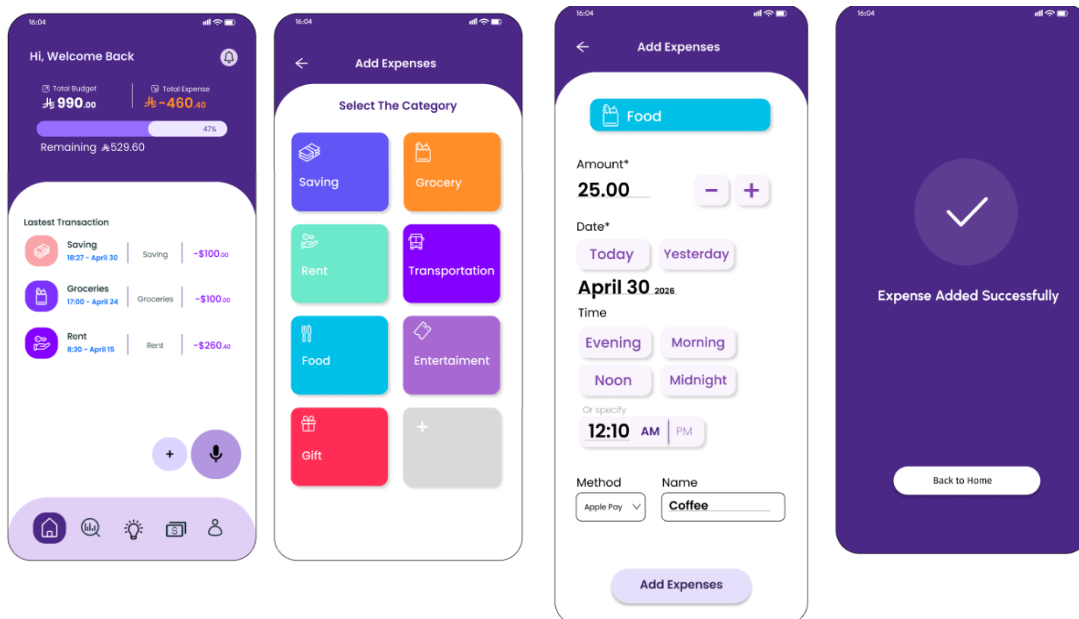


PROTOTYPE EXPLANATION

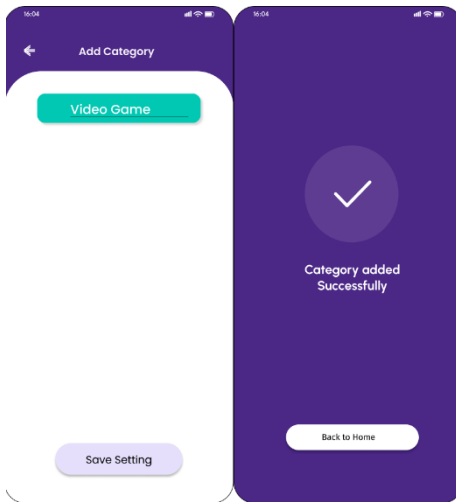
1 log-in/sign-up



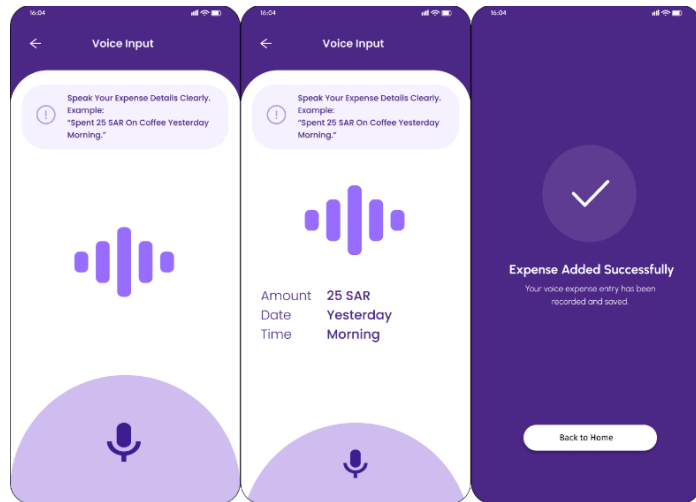
2 Home And add expenses



2.1 Add category



3 Add expense using voice command

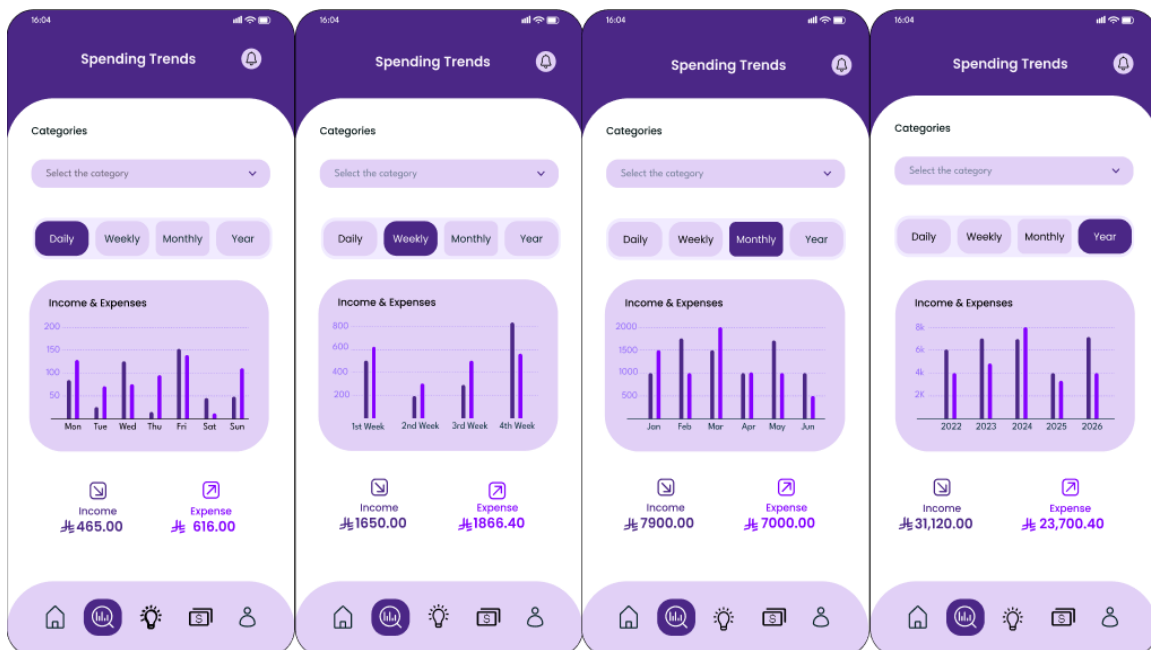


This section allows users to quickly add and track their daily expenses through a simple and organized interface. Users can manually add expenses or use the voice input feature to record expenses more quickly. The system allows users to select spending categories, enter expense details such as amount, date, and payment method, and save transactions successfully within the application.

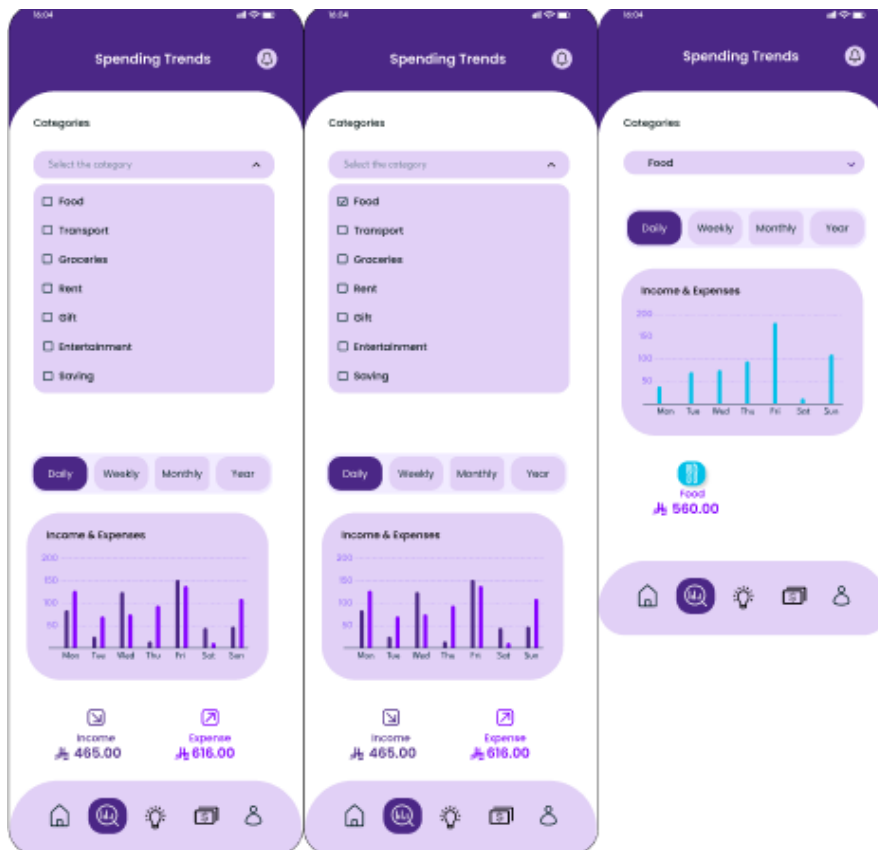
The voice input feature was included to provide a faster and more convenient expense entry experience for students. Users can verbally describe their expense details, and the system records the information automatically. Manual expense entry is also available for users who prefer traditional input methods.

The design goals for this section were to simplify the expense entry process, reduce cognitive load using categorized expense options, support flexible interaction through both manual and voice input, maintain clear navigation between screens, and improve financial tracking through simple forms and confirmation feedback.

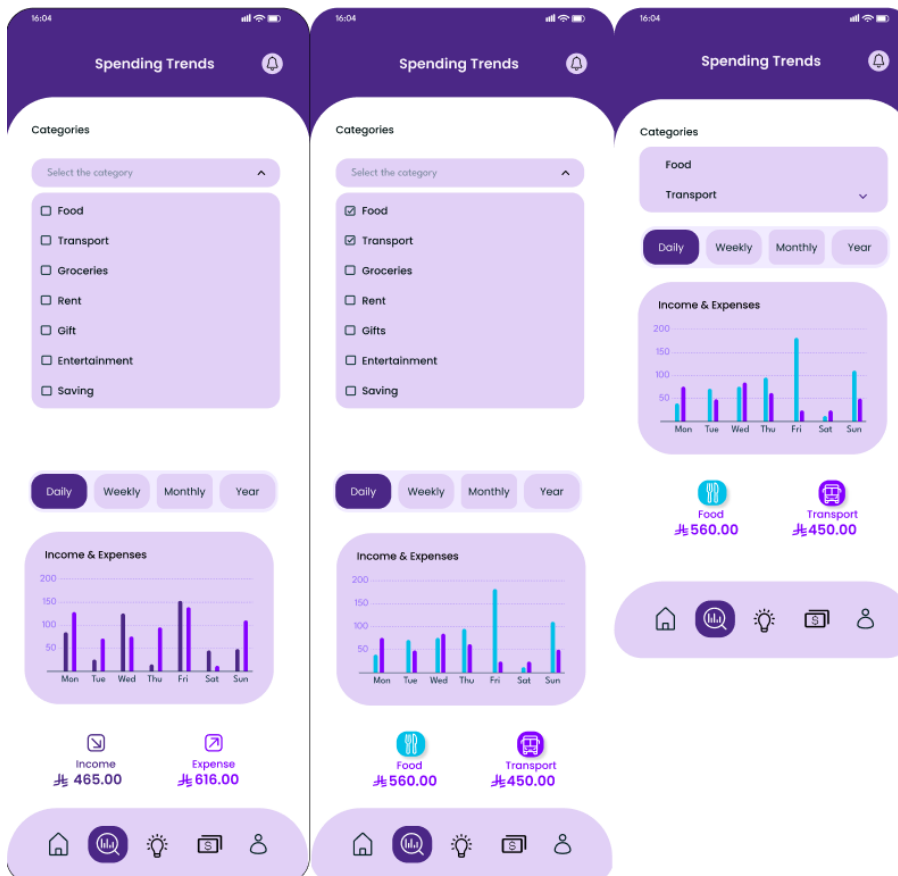
4 Spending Trends



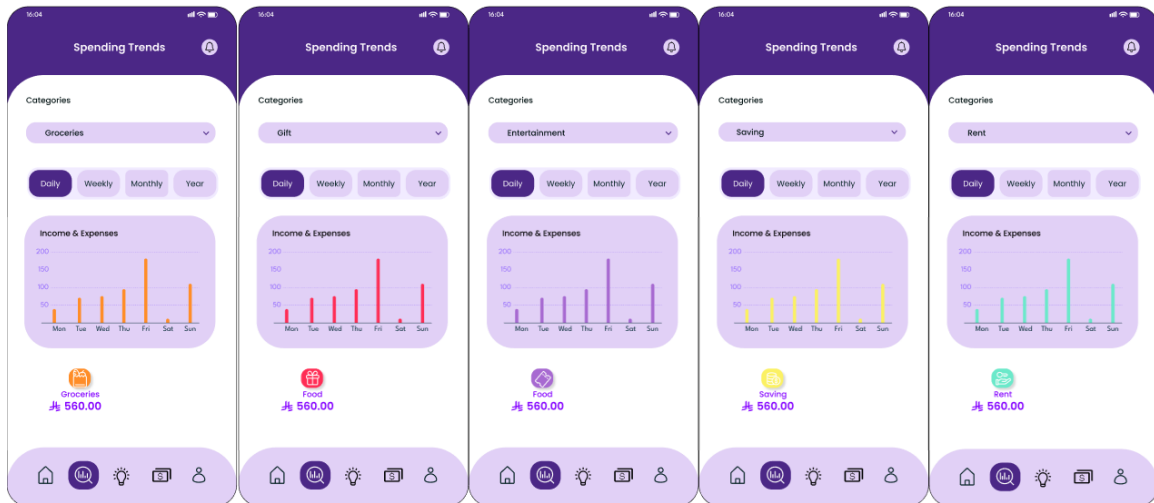
4.1 Spending Trends for food



4.2 Spending Trends for food and transcation



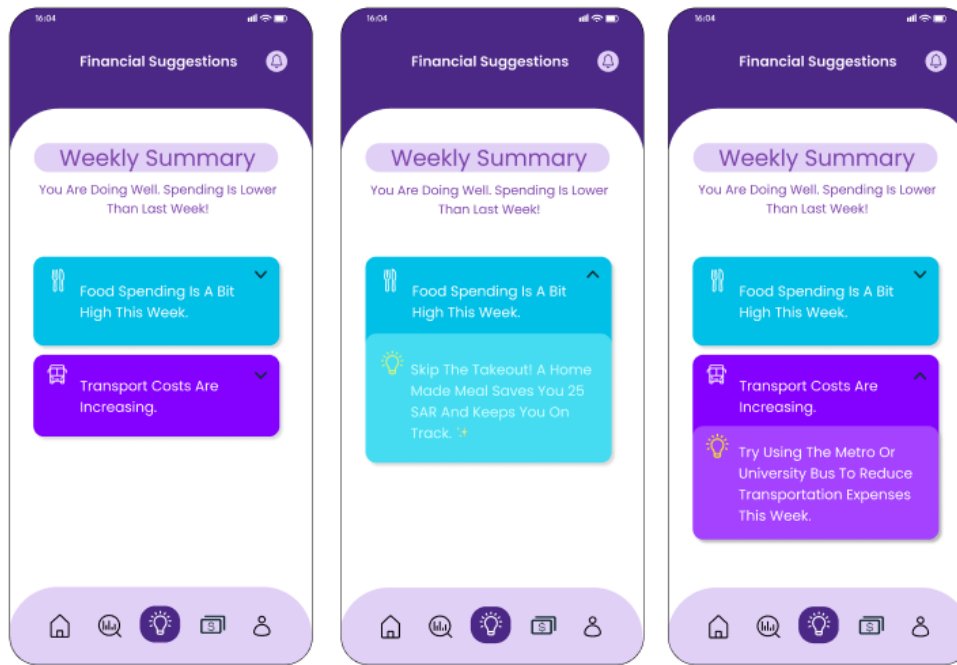
4.3 Spending Trends for every category



This section helps users understand their spending behavior through visual charts and category analysis. Users can view their expenses based on different time periods such as daily, weekly, monthly, and yearly trends. The system also allows users to filter spending categories and identify areas with higher expenses.

The design goals for this section were to present financial information in a clear and visually simple way, reduce cognitive effort using charts and color indicators, help students identify overspending categories quickly, and support better financial decision-making through spending insights and trend visualization.

5 Financial Suggestions

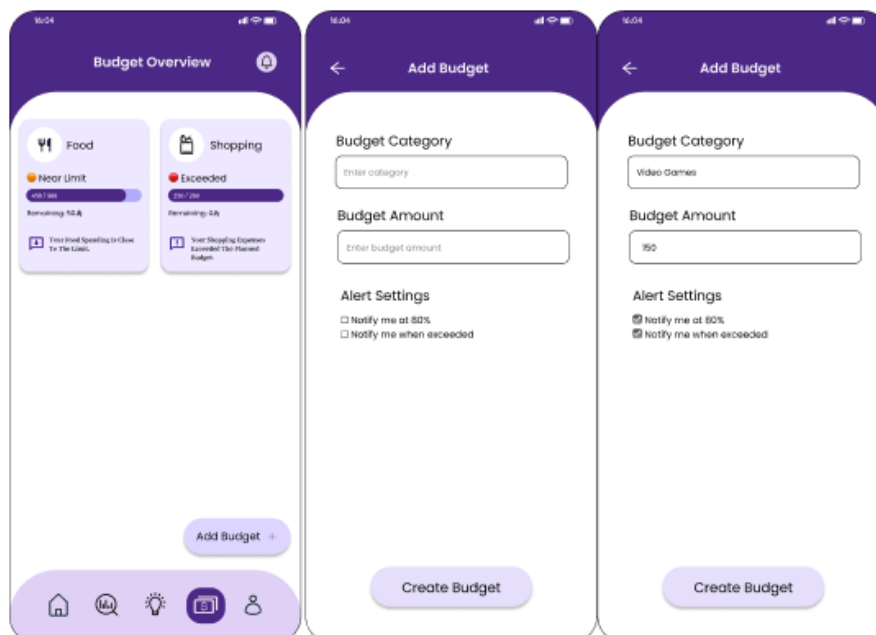


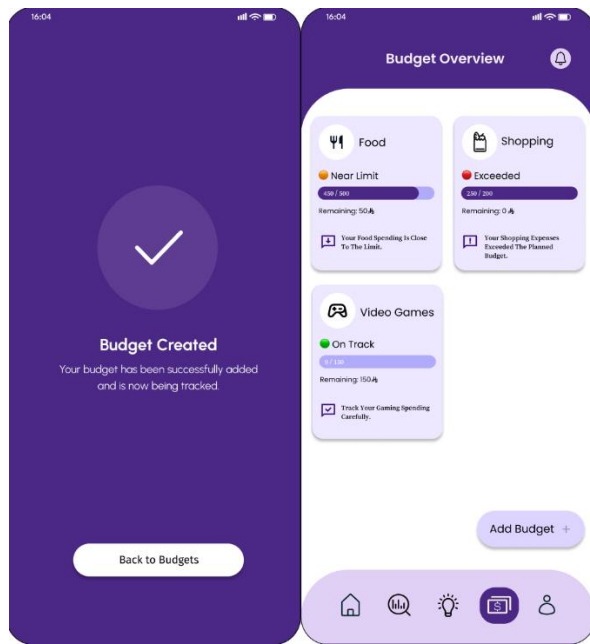
This section provides users with personalized financial suggestions based on their spending behavior and budget usage. The system analyzes categories with higher expenses and offers simple recommendations to help students manage their allowance more effectively.

Users can view weekly financial summaries, identify categories with increased spending, and receive practical suggestions for reducing unnecessary expenses. Interactive expandable cards were used to keep the interface clean while allowing users to view more detailed advice when needed.

The design goals for this section were to support better financial decision-making, provide friendly and motivating feedback, reduce financial stress for students, and present suggestions in a simple and easy-to-understand format.

6 Budget



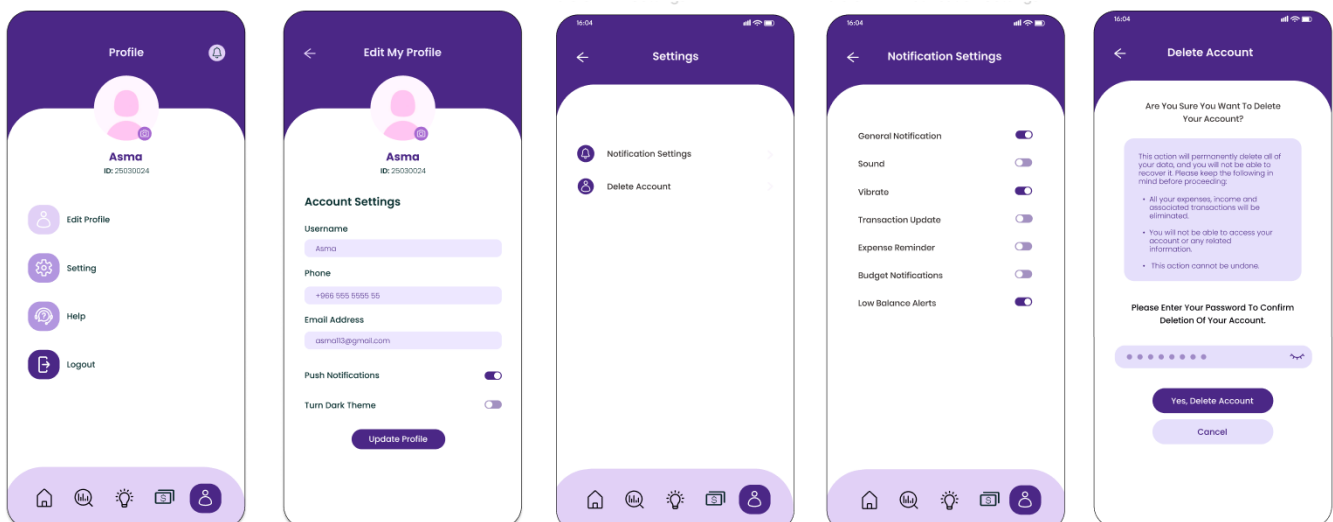


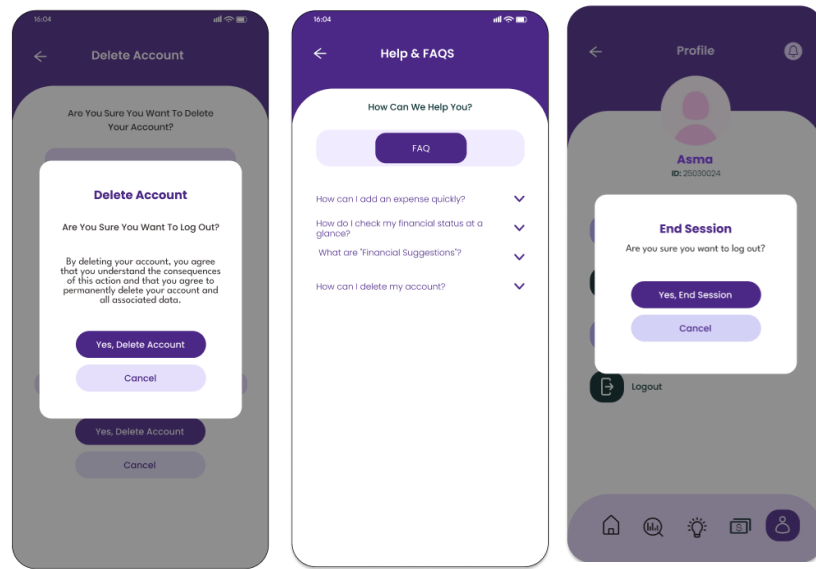
This section allows users to create and manage spending budgets for different expense categories such as food, shopping, and video games. Users can set budget amounts, enable alert notifications, and monitor spending progress through visual budget cards and progress indicators.

The system provides spending status labels such as “On Track,” “Near Limit,” and “Exceeded” to help students quickly understand their financial situation. Users can also add new budgets and receive confirmation feedback after successful budget creation.

The design goals for this section were to help students manage their limited allowance more effectively, reduce overspending through budget alerts, simplify budget tracking using visual indicators and progress bars, and support financial awareness through clear and organized budget information.

7 Profile





This section allows users to manage their personal account information and application settings. Users can edit profile details, control notification preferences, access help and FAQ pages, and manage account-related settings such as logging out or deleting the account.

The profile interface was designed using a clean and organized layout to make navigation simple and accessible for students. Toggle settings and clear account management options were used to improve usability and reduce user effort during interaction.

The design goals for this section were to provide users with easy access to account management features, maintain clear and organized navigation, improve personalization through editable profile settings, and support user control and accessibility within the application.

USABILITY TESTING RESULTS

Testing Method

The usability test was conducted using an interactive Figma prototype. Participants were asked to complete several common tasks related to budget management and expense tracking while their interactions and feedback were observed.

Testing Tasks

1. Add a new budget
2. Check budget progress in the Budget Overview screen
3. Open a budget alert notification
4. Add a new expense
5. View spending trends
6. Open the Financial Suggestions screen

Testing Duration

- Participant 1: 12 minutes
- Participant 2: 10 minutes

Participant 1 Feedback

- The participant experienced confusion while trying to add an expense and create a budget at the same time.
- The participant expected the system to allow budget creation during the expense entry process instead of requiring separate steps.
- The participant was unsure whether the expense could be tracked correctly without first creating a budget.

Participant 2 Feedback

- The participant did not understand the purpose of the microphone icon on the home screen.
- The participant was unsure that the feature was intended for voice expense entry.

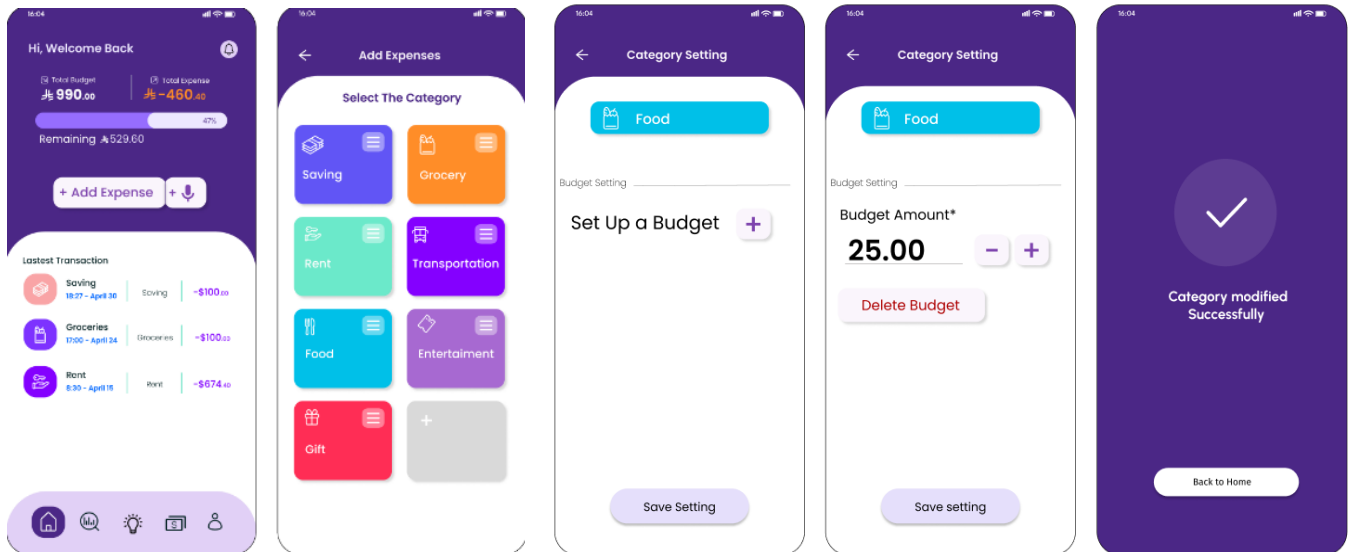
IMPROVEMENTS

Improvement 1

Based on user feedback, the expense entry flow was redesigned to reduce confusion when adding expenses and budgets together, when creating a new category or when clicking on the three-line button to open the category setting.

Users can now:

- Create a budget while creating a category
- Create a budget from the category settings
- Continue without creating a budget



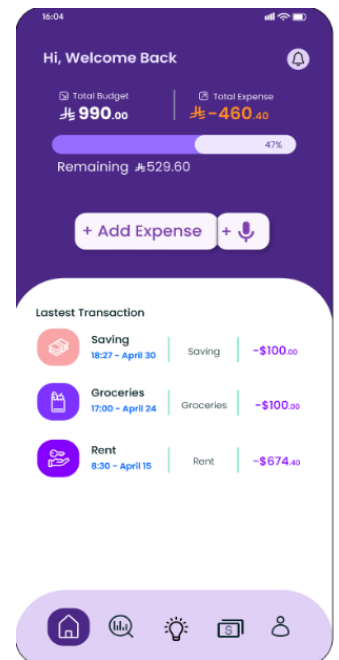
Improvement 2

Based on user feedback, the microphone shortcut was replaced with a clearer

“Add Expense” button. Users can now choose whether they want to:

- Add expenses using voice input
- Add expenses manually

This made the feature easier to understand and improved navigation clarity.



APPENDIX 1 – QUESTIONNAIRE QUESTIONS

1. Are you a university student?

- Yes
- No
- Graduate
- Job seeker

2. What is your age?

- Under 18
- 18–25
- 26–35
- 35+

3. Do you use an expense-tracking app?

- Yes
- No

4. How often do you track your expenses?

- Never
- Monthly
- Weekly
- Daily

5. Manually entering expenses is tiring. *(Likert)*

Strongly disagree / Disagree / Neutral / Agree / Strongly agree

6. I sometimes forget to log my expenses. *(Likert)*

Strongly disagree / Disagree / Neutral / Agree / Strongly agree

7. I feel confused when looking at financial data or charts. *(Likert)*

Strongly disagree / Disagree / Neutral / Agree / Strongly agree

8. I would like the app to give me suggestions on how to spend my money better. *(Likert)*

Strongly disagree / Disagree / Neutral / Agree / Strongly agree

9. What problems do you face when tracking your expenses? *(open-ended free text)*

10. Adding a voice-input feature (example: the app records "I spent 15 riyals on groceries today" and categorises it) would make logging expenses easier for me. *(Likert)*

Strongly disagree / Disagree / Neutral / Agree / Strongly agree

11. If you could choose one additional feature, which would you pick?

- Budget alerts (notifications when approaching or exceeding the limit)
- Subscription detection (identifying recurring payments or subscriptions)
- Savings-goal tracking (setting and following savings targets)
- Shared / family budget (managing expenses with others)

APPENDIX 2 – RESPONSES OF THE OPENED QUESTION (QUESTION 9)

١. ادخال البيانات ياخذ وقت طويل جدا وبعض التطبيقات لا تعطي نصائح لتقليل المصاريف
٢. خاصية الادخال الصوتي لديكم مفيدة اذا دعمتوا التطبيق يكون للاطفال بحيث يحفزهم عالادخار
٣. أنسى القيام بتببع النفقات النقدية أحيانا / صعوبة التتبع البنكي و النقدي بدقة
٤. الاعتماد على الذاكرة وعدم تسجيلها أول بأول
٥. تذكر محل او جهة الشراء، معرفة مجموع الانفاق الشهري، تصنيف النفقات
٦. المصاريف العائليه.
٧. كثرتها
٨. أنى أنسى أسجلها يوميا
٩. تسجيلها واحدة واحدة
١٠. أنسى اكتبه ، م امشي على الخطه
١١. أنسى وش اللي صرفت عليه
١٢. اضافة النفقات يدويا متعب وياخذ وقت وانسى
١٣. ممكن مشكلة التقسيم الأموال ووضع ميزانية مناسبة النفقات مثل نفقة الفواتير او مثلا الترفيه وأخرى وتقسيما ع حسب مثلا عدد أفراد الأسرة وتفضيلاتهم يعني ممكن احد الأفراد يفضل مطعم معين او مكان فيجب أخذها ب
١٤. اكره لما ينقلني لشهر جديد قبل اسجل نفقات الشهر اللي قبله وكمان ماعنده خيارات كافية وواضح عن الاشياء اللي اصرف عليها زي مثلا أنا صارفه على نفسي مفروض يحط فئات تحتها هل منتجات عنابه فيتامينات عشان اعرف
١٥. معرفة تواريخ استحقاق التزاماتي المالية الشهرية وهل المتبقي في الحساب كافي لتغطية هذه الالتزامات حتى نهاية الشهر
١٦. كل يوم اشترى اشياء سواء قهاوي واكل في الجامعة او اونلاين اطلب اشياء، وما احس فيه وقت اني اجلس ادخل كل شيء شريته. احس فيه خاصية حلوة لوسويتوها بتصبرون خرافيين، اربطوا تطبيقكم بابل باي لان الغالب يدفع من
١٧. لدي عدة بطاقات ويصعب علي متابعتها ومعرفة المصروفات بالضبط
١٨. عدم المتابعه بشكل دوري والنسيان

FIGMA URL

Figma URL: <https://www.figma.com/design/p6NyiAGfVrK5wByZulWBPy/HCI?node-id=0-1&t=IA7JKCGtMRoM8Cgk-1>

Figma interactive prototype: <https://www.figma.com/proto/p6NyiAGfVrK5wByZulWBPy/HCI?node-id=557-1692&p=f&t=PaAWlhTw8cLN7Yhe-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=53%3A841&show-proto-sidebar=1>

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